

OZONE DENSITY DISTRIBUTION IN THE MESOSPHERE (50-90 KM) MEASURED BY THE SME
LIMB SCANNING NEAR INFRARED SPECTROMETER

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Abstract. The ozone densities between 50 and 90 km are deduced from 1.27 μm airglow measured on the Solar Mesosphere Explorer satellite. The derived densities agree well with those made simultaneously from SME by the ultraviolet spectrometer. The data set extends from pole to pole at about 3 pm, for most sunlit latitudes. At low altitudes, in the mesosphere, there are larger variations in ozone density in the winter latitudes than in the summer. Above the mesopause the day-to-day variation in ozone density is a factor of two at most latitudes and times.

Introduction

The Solar Mesosphere Explorer (SME) spacecraft was launched on October 6, 1981 into a nearly circular, sun-synchronous orbit, so that the subsatellite local time is near 3:00 pm or 3:00 am. The scientific objectives of SME include studies of ozone photochemistry in the mesosphere. The combination of spacecraft instruments is designed to measure ozone, its variations in time and space, and correlate these to changes in the relevant solar flux, the temperature, and other important atmospheric constituents, e.g. water vapor (Barth et al., 1983).

The near infrared instrument measures emission from $\text{O}_2(^1\Delta_g)$ at 1.27 μm . In the dayglow the oxygen molecule is excited to this state primarily by photodissociation of ozone and emits at 1.27 and 1.58 μm . The emission at 1.58 μm was first observed from the ground in the evening twilight by Vallance Jones and Harrison (1958). The emission at 1.27 μm is stronger, but is absorbed in the troposphere by molecular oxygen. Vallance Jones and Gattinger (1963) proposed that the emission was due to ozone photodissociation. A rocket profile of the dayglow was made by Evans et al. (1968). Their analysis indicated that the emission was due to ozone photodissociation, except possibly above 80 km where the excess emission required either more ozone than predicted at that time, or another source of $\text{O}_2(^1\Delta_g)$ molecules. Our analysis indicates that both are true; there is, at times, a secondary peak of ozone density near 85 km, while near 80 km often only about half the emission is due to ozone.

In this paper we report on initial results from the near infrared spectrometer on SME which measures ozone in the 50-90 km region. We will discuss the methods that are used to invert the observed intensities giving the volume emission

rate. The ozone densities are then inferred assuming several sources for the emission. The general behavior of the ozone with altitude, latitude, and time will be shown and discussed.

Observations and Data Analysis

The near infrared instrument on SME is a two-channel Ebert-Fastie spectrometer with a programmable wavelength drive. The collecting optics are an f/5 25 cm focal length off-axis parabolic telescope which feeds an f/5 12.5 cm focal length Ebert-Fastie spectrometer. The field of view is $0.074^\circ \times 0.74^\circ$ which projects to 3.5 km X 35 km at the earth's limb. The instrument employs a 200 groove/mm grating blazed at 1.8 μm and two passively cooled lead sulfide detectors. The spectral resolution is 12.25 nm and the channels are separated by 0.6 μm . The wavelength coverage for channel 1 and channel 2 are 1.1-2.6 μm and 1.1-3.2 μm respectively. The grating may be operated either in a scanning mode with 44 nm steps or commanded to any of 512 positions. The data reported here were obtained on the Earth's limbs with a fixed grating position at 1.27 μm .

The SME spacecraft spins in cartwheel fashion once every 12 seconds with the spin axis perpendicular to the orbit plane. Thus, one scan of the Earth's limb is accomplished every 12 seconds, or about 1 degree of latitude along the orbit path. During each scan of the Earth's limb, 31 data samples are collected. The integration period of 2.44 ms gives 3.5 km resolution on the limb and 31 samples cover the altitude region from about 30 to 130 km. Most of the data were taken on the orbits that cover the longitude range from 40 to 100 degrees west each day during local afternoon.

The altitude profile of typical radiance data is shown in Figure 1. The radiance measurements taken on the limb forward of the spacecraft in the orbit plane begin at the highest altitude with a measurement of the dark sky background. The signal increases above the noise between 90 and 100 km. Often an inflection or even a relative minimum in the signal is seen near 80 km. The signal reaches a peak near 45 km and then falls off until the Rayleigh scattering from the lower atmosphere, scattering from clouds, and reflection from the earth's surface are seen. During the day, the maximum airglow signal is between 250 and 350 data numbers or DN (600 and 800 MR), while at night the maximum signal is about 1 DN (2.3 MR). The instrument sensitivity is about 2 MR per data number and the noise on one sample is about 1.5 DN. Thus, the day signals have a very good signal-to-noise ratio while at night it is poor. The 8-bit data compression adds an error of up to $\pm 3\%$ to all data points.

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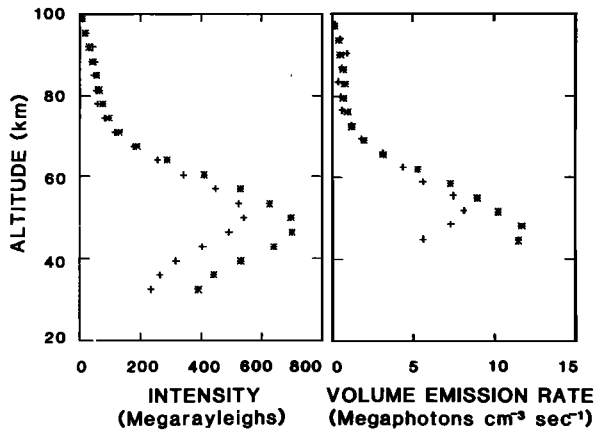


Figure 1: Data measured by the limb scanning near infrared spectrometer on SME on 14 March, 1982 on the left. These profiles are six data scans that have been merged and calibrated. The inverted data, $1.27 \mu\text{m}$ volume emission rate, is on the right. (+) -data at 66.2°S latitude and a solar zenith angle = 74.1° . (*) -data at 12.5°N and a solar zenith angle = 44.1° .

The data are calibrated by taking into account the instrument and detector temperatures which are included in the telemetry. In order to decrease the noise and the volume of data storage, we merge 6 calibrated profiles together taking into account any changes in pointing and timing.

The observed slant intensity along the line of sight path l with a tangent altitude of h is

$$S(h) = \int_0^\infty V(l)T(l)dl$$

Where $V(l)$ is the volume emission rate. The $1.27 \mu\text{m}$ emission is self absorbed by O_2 . $T(l)$ is the atmospheric transmission of $1.27 \mu\text{m}$ radiance between the emission point and SME and depends on the column density of O_2 and the emission and absorption temperatures.

The volume emission rate $V(h)$, at any altitude h , is retrieved using an onion peel method assuming that the atmosphere is uniformly stratified. Two inverted profiles are shown in Figure 1.

During the day, the dominant production mechanisms of $\text{O}_2(^1\Delta)$ are ozone photodissociation and collisional transfer of energy from $\text{O}_2(^1\Sigma)$. We examined the photochemical reactions (using cur-

TABLE 1 REACTION RATES AND CONSTANTS

ϵ	0.9	1
J_3	$8.0 \times 10^{-3} \text{ sec}^{-1}$	2
$J_{s\text{O}}$	$6.3 \times 10^{-9} \text{ sec}^{-1}$	3
$A_{s\text{O}}$	0.085 sec^{-1}	3
A_D	$2.58 \times 10^{-4} \text{ sec}^{-1}$	4
k_D	$2.22 \times 10^{-18} (T/300)^{0.78} \text{ cm}^3 \text{ sec}^{-1}$	5
k_n	$2.0 \times 10^{-11} \exp(-107/T) \text{ cm}^3 \text{ sec}^{-1}$	6
k_o	$2.9 \times 10^{-11} \exp(-67/T) \text{ cm}^3 \text{ sec}^{-1}$	6
k_s	$2.0 \times 10^{-15} \text{ cm}^3 \text{ sec}^{-1}$	7

1. Fairchild et al. 1978, 2. See text, 3. Wallace and Hunten 1968, 4. Badger et al. 1965, 5. Findlay and Snelling 1971, 6. Streit et al. 1976, 7. Becker et al. 1971.

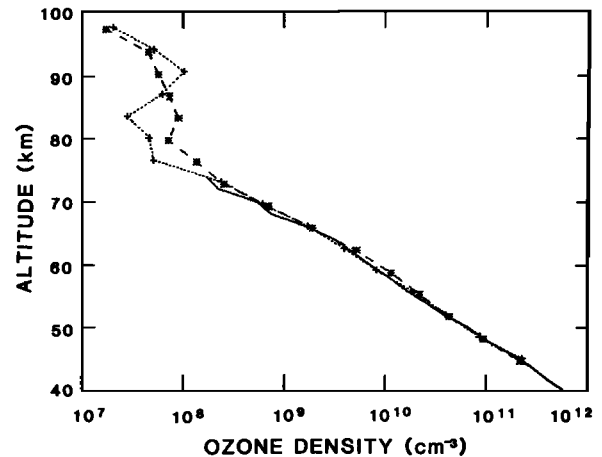
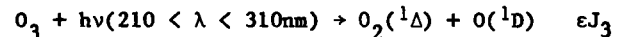
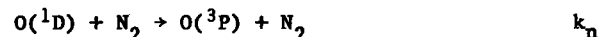
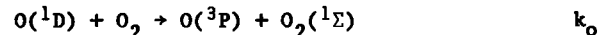


Figure 2: Ozone density at 65°S and 10°N corresponding to the profiles of Figure 1. The solid line ending at 72 km is the average data of Krueger and Minzner (1967).

rent rate constants) that could produce $\text{O}_2(^1\Delta)$ (Evans et al., 1968 and Han et al. 1973) and found only those discussed below were significant during the day.



The production efficiency ϵ is about 90% and the number of ozone dissociations per second per molecule is J_3 . J_3 is calculated using temperature dependent cross sections of Bass and Paur (1982) and the solar flux of Mount and Rottman (1983) corrected for the sun-earth distance and the overhead ozone. The excited oxygen atom can transfer its energy to O_2 or be quenched by N_2 .



The $\text{O}_2(^1\Sigma)$ molecule in turn is de-excited by

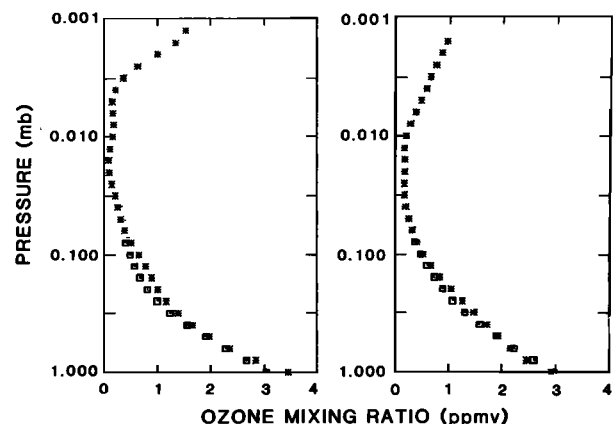


Figure 3: The same ozone densities as in Figure 2 but plotted as mixing ratio in parts per million by volume vs. pressure. 65°S is on the left and 10°N is on the right. The small squares are ozone data from the UV absorption experiment (Rusch et al., 1983).

where M is any molecule, usually N_2 . All de-excitation reactions of $O_2(^1\Sigma)$ produce an $O_2(^1\Delta)$ (Ogryzlo and Thrush, 1974). Deactivation of $O_2(^1\Sigma)$ effectively adds 30% to the photodissociation production of $O_2(^1\Delta)$, making the efficiency about 117%. $O_2(^1\Sigma)$ is also produced by resonance absorption of sunlight at 762nm which is then collisionally converted to $O_2(^1\Delta)$ below about 90 km. Above that altitude more $O_2(^1\Sigma)$ is lost by radiation than by quenching. The excitation rate is J_s .

$O_2(^1\Delta)$ molecules either radiate or are quenched by O_2



Hence the volume emission rate is given by

$$V(h) = \left\{ \frac{P_s k_s [M]}{A_s + k_s [M]} + J_3 \epsilon [O_3] \right\} \frac{A_D}{A_D + k_D [O_2]}$$

A_s and A_D are the Einstein coefficients for $O_2(^1\Sigma)$ and $O_2(^1\Delta)$ respectively, and P_s is the production rate of $O_2(^1\Sigma)$

$$P_s = J_s [O_2] + J_3 \epsilon [O_3] K_o / (K_o + 3.76 K_n)$$

The above equations are then solved for the ozone density and mixing ratio. The background atmospheric density and temperature are taken from Cole and Kantor (1978). Rates and constants used are given in Table 1. The transmission function for resonance-band absorption by O_2 molecules was taken to be a weighted-sum over all lines of the transmission function for a pure Doppler line (Holstein, 1947). The weighting

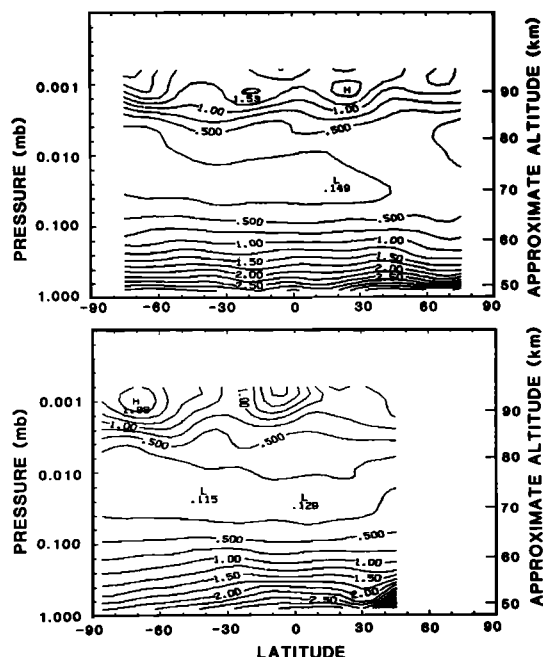


Figure 4: Contour plots of ozone mixing ratio for the orbit of 14 March, 1982, top, and 22 December, 1981 on the bottom. The upper plot contains data shown in Figure 3. Negative value of latitude indicates the southern hemisphere.

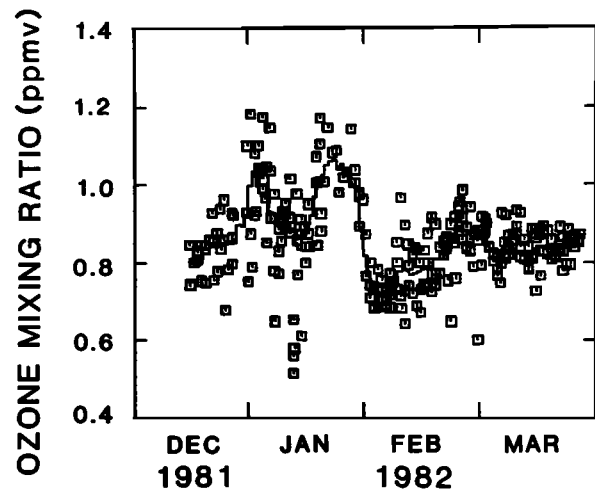


Figure 5: Ozone mixing ratio as a function of time at 40°N and about 60 km (0.159 mb). Each square is one orbit while the solid line is a five day running average.

factor is the normalized Boltzmann distribution function evaluated at the temperature of the model atmosphere evaluated at the limb point (Wallace and Hunten, 1968).

Results and Discussion

The ozone densities and mixing ratios derived from the intensity measurements of Figure 1 are shown in Figures 2 and 3. In Figure 2 the logarithm of the ozone density is plotted against altitude. Below 70 km the ozone follows an exponential behavior and shows a good agreement with the rocket model of Krueger and Minzer (1976). Above 70 km the scale height increases and often there is a secondary maximum of the ozone density. The mixing ratios are shown in Figure 3 with total pressure as the vertical coordinate. The profiles are representative of March conditions. Figure 3 also includes data from the SME UV measurement of ozone that shows the good agreement of ozone altitude profiles.

The ozone inferred from the measurements exhibits a considerable variation with latitude and time. The latitude variation of the ozone mixing ratio by volume for two orbits is represented by contour maps in Figure 4. In the winter, the mixing ratio is enhanced in the northern mid latitudes. Above 70 km (0.03 mb), there is a good deal of structure in each map. The mixing ratio above 80 km has a maximum near the summer pole.

The variation of ozone with time can be seen in Figure 5 for 40°N latitude at 60 km. The changes from orbit to orbit are due to longitudinal variations and random noise in the inversion. In Figure 6, the 5 day running average is shown for several pressure levels at 40°S. At high pressures (1.0 and 0.8 mb) ozone mixing ratio increases throughout the period up to the beginning of May when large winter variability becomes apparent. The dotted line at 0.8 mb represents ozone measured by the UV spectrometer. The agreement between the two instruments is very good. The ozone at 0.3, 0.1, and 0.01 mb

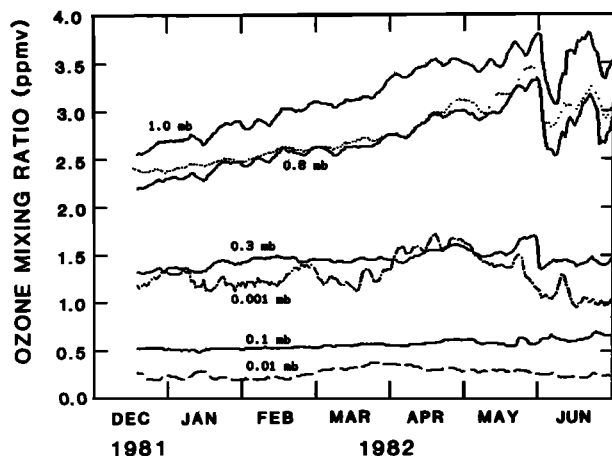


Figure 6: Ozone five day running average at several pressure levels at 40°S.

appears less variable except in June. The variability returns at 0.001 mb throughout the entire period, possibly indicating a region of control of ozone different than that at higher pressures.

Satellite measurements of the 1.27 μm airglow on the Earth's limb are currently being made from the SME spacecraft. Most of the emission is due to ozone photodissociation. A small amount of the emission is due to transfer of energy from $\text{O}_2(^1\Sigma)$ which is resonantly excited by sunlight at 762 nm. Taking into account quenching of both $\text{O}_2(^1\Sigma)$ and $\text{O}_2(^1\Delta)$, the atmospheric transmission of the UV, of 762 nm and of 1.27 μm radiation, we infer the ozone densities from 50 to 90 km during the daylight portion of the orbit.

We have shown that the derived ozone density is consistent with that measured previously by rockets and with the UV absorption measurements of ozone on SME. The ozone above 80 km is highly variable while that at 60 km shows seasonal trends and winter variability.

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